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CS 621 DATA ANALYTICS

Data Analytics Project Planning

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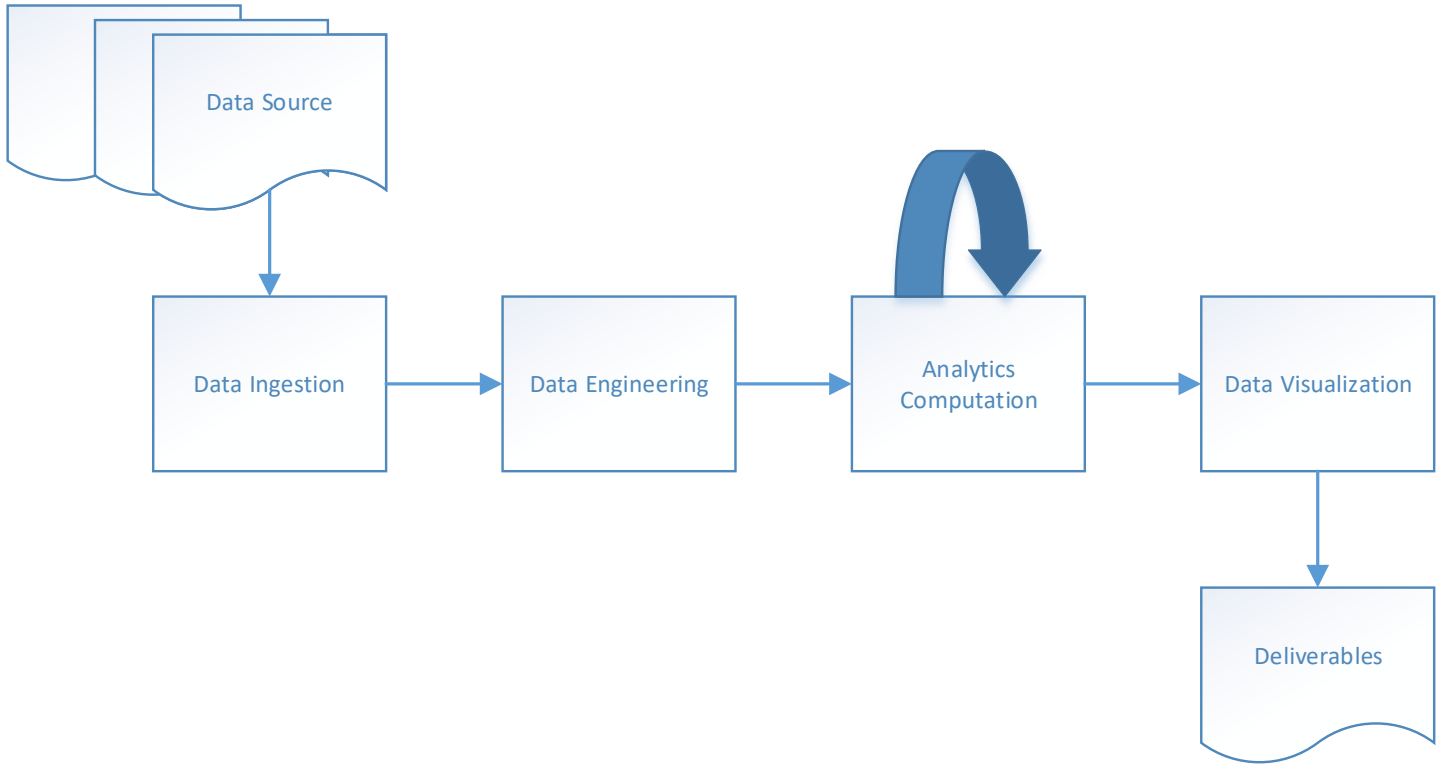
DATA ANALYTICS PROJECT PLAN

- Plan for a data analytics project:
  - Domain and application
  - Data availability (i.e. data source)
  - Essential work estimation
    - Data ingestion
    - Data engineering
    - Data analytics computation
    - Data visualization
  - What to deliver?

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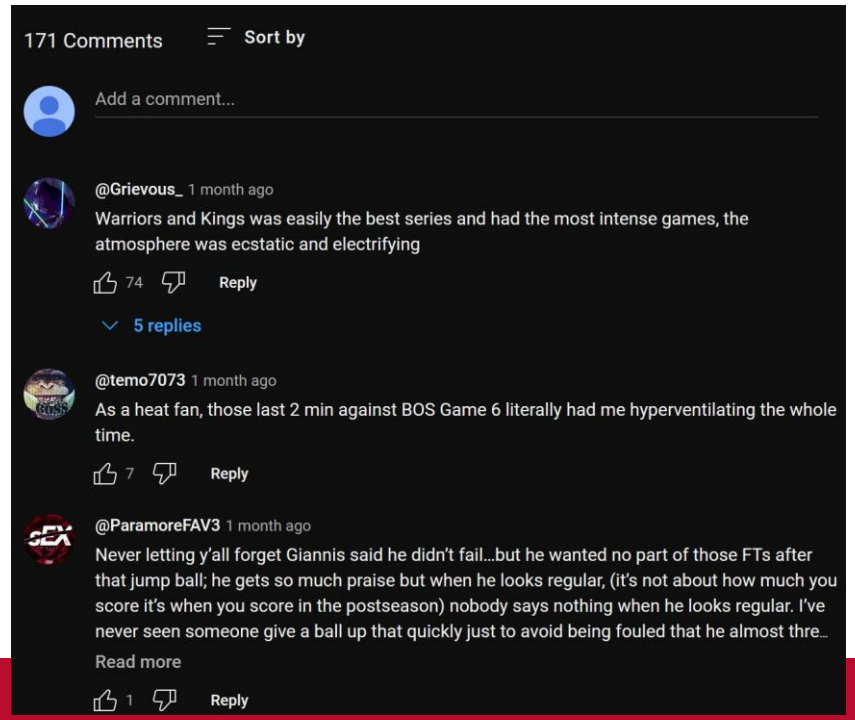
BLUEPRINT

- Follow the data science pipeline as a blueprint.



DOMAIN AND APPLICATION

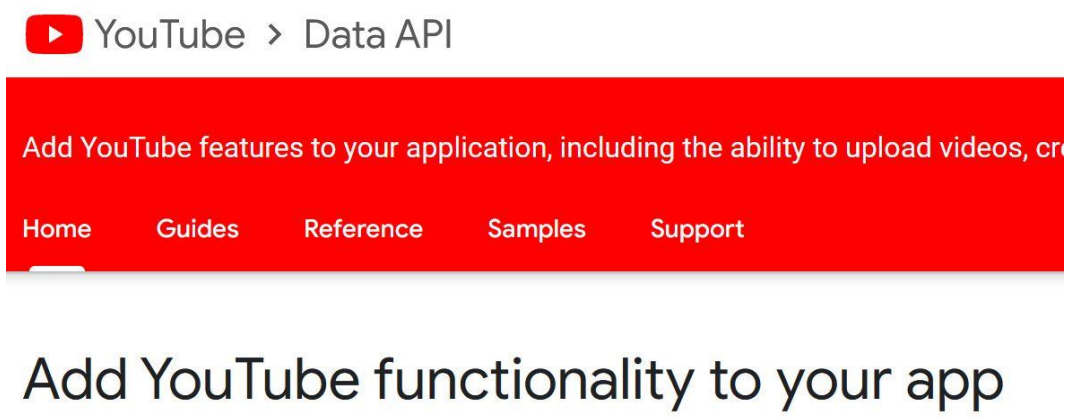
- Application domain:
  - Social Network
  - Youtube viewer comments analysis on popular influencers.



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DATA SOURCE

- Data availability:
  - Retrieve data via YouTube Data API



  - plan to collect 10,000 comments from 10 video clips by 2 influencers.
  - hours needed: 10 hours.
  - Raw data collected will be stored in json files.

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DATA INGESTION

- Data collected from YouTube Data API has been stored in local JSON files.
- After loading data from local files into Python programs, the following data structures will be used to store the raw data:
  - Lists
  - Dictionaries
  - Pandas Series
  - Pandas DataFrame
  - hours needed: 5 hours.

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DATA ENGINEERING

- In this stage, the following work will be done:
- Removal of
  - empty comments
  - comments written in different languages
  - duplicated comments
  - outdated comments
- Hours needed: 8 hours

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ANALYTICS COMPUTATION

- Basic analytics:
  - # of comments
  - # of empty comments
  - # of reviewers
  - lengths of comments
  - …
- Hours needed: 10 hours

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ANALYTICS COMPUTATION

- Advanced analytics
  - Comments polarity; Sentiment Analysis
  - Comments analyses as time series data
  - Comments analyses as spatial data
  - Emoji emotion analysis
  - Comparisons between influencers
- Hours needed: 10 hours

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VISUALIZATIONS

- When applicable, provide
  - Boxplots
  - Bar charts
  - Pie charts
  - Scatter plots
  - Time Series visualization
  - Spatial visualization (optional, geo-tagging may not be available)
  - Wordclouds
- Hours needed: 12 hours

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DELIVERABLES

- Raw data and cleaned dataset (~10,000 and ~6000 rows)
- A written report (6-10 pages)
- A 10-12 pages PPT presentation (15 minutes)
- A Python program for interactive demonstration (~300 lines of code with documentation)

- Hours needed: 10 hours



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WE FLY